Derivation of FM/PM signals

A. For FM signals,

$$f_{i,FM}(t) = f_c + k_f V_m \cos(2\pi f_m t)$$

$$\omega_i(t) = 2\pi f_i(t)$$

$$\frac{d\theta_i(t)}{dt} = \omega_i(t) = 2\pi\left[f_c + k_f V_m \cos(2\pi f_m t)\right]$$

$$\begin{aligned}
\theta_i(t) &= \int_0^t 2\pi\left[f_c + k_f V_m \cos(2\pi f_m \tau)\right] d\tau \\
&= \int_0^t 2\pi f_c d\tau + \int_0^t 2\pi k_f V_m \cos(2\pi f_m \tau) d\tau \\
&= 2\pi f_c t + 2\pi k_f V_m \int_0^t \cos(2\pi f_m \tau) d\tau \\
&= 2\pi f_c t + \frac{2\pi k_f V_m}{2\pi f_m} 2\int_0^t \cos(2\pi f_m \tau) d(2\pi f_m \tau) \\
&= 2\pi f_c t + \frac{k_f V_m}{f_m}\sin(2\pi f_m t)
\end{aligned}$$

$$\therefore \quad v_{FM}(t) = V_c \cos\left[\theta_i(t)\right] = V_c \cos\left[2\pi f_c t + \frac{k_f V_m}{f_m}\sin(2\pi f_m t)\right]$$

let $\beta = \dfrac{k_f V_m}{f_m}$, then $v_{FM}(t) = V_c \cos\left[2\pi f_c t + \beta \sin(2\pi f_m t)\right]$

thus, $f_{d,FM} = \frac{1}{2} f_{p-p} = \frac{1}{2}\left(f_{i,\max} - f_{i,\min}\right) = \frac{1}{2}\left[\left(f_c + k_f V_m\right) - \left(f_c - k_f V_m\right)\right] = k_f V_m$

Therefore, $f_{d,FM}$ is proportional to V_m, and is independent of f_m.

B. For PM signals,

$$\theta_{i,PM}(t) = 2\pi f_c t + k_p V_m \cos(2\pi f_m t)$$

$$\therefore v_{PM}(t) = V_c \cos\left[\theta_i(t)\right] = V_c \cos\left[2\pi f_c t + k_p V_m \cos(2\pi f_m t)\right]$$

$$\begin{aligned}
f_{i,PM}(t) &= \frac{1}{2\pi}\frac{d\theta_{i,PM}(t)}{dt} \\
&= \frac{1}{2\pi}\left[2\pi f_c - 2\pi f_m k_p V_m \sin(2\pi f_m t)\right] \\
&= f_c - f_m k_p V_m \sin(2\pi f_m t)
\end{aligned}$$

thus, $f_{d,PM} = \frac{1}{2} f_{p-p} = \frac{1}{2}\left(f_{i,\max} - f_{i,\min}\right) = k_p f_m V_m$

Therefore, $f_{d,PM}$ is proportional to both V_m, and f_m.

Bessel functions

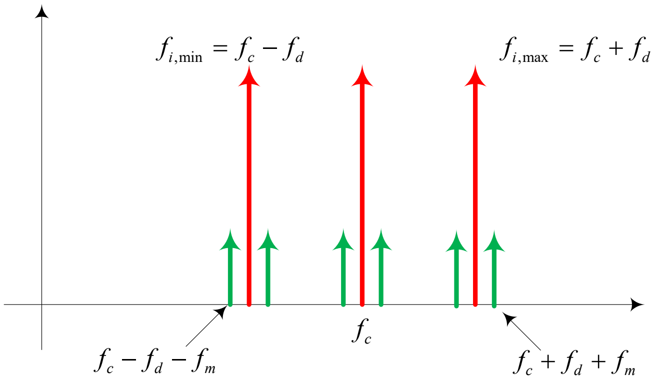
Three Bessel function identities:

$$\cos(z\sin(\theta)) = J_0(z) + 2\sum_{n=1}^{\infty} J_{2n}(z)\cos(2n\theta)$$

$$\sin(z\sin(\theta)) = 2\sum_{n=0}^{\infty} J_{2n+1}(z)\sin\left[(2n+1)\theta\right]$$

$$J_{-n}(z) = (-1)^n J_n(z)$$

$$V_c \cos\left[2\pi f_c t + \beta \sin\left(2\pi f_m t\right)\right] = V_c \sum_{n=-\infty}^{\infty} J_n\left(\beta\right) \cos\left[2\pi\left(f_c + n f_m\right) t\right]$$



$$BW_{Carson} = \left(f_c + f_d + f_m\right) - \left(f_c - f_d - f_m\right) = 2\left(f_d + f_m\right)$$